

Bitwuzla at the SMT-COMP 2026

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Abstract—In this paper, we present our Satisfiability Modulo Theories (SMT) Bitwuzla as submitted to SMT-COMP 2026. Bitwuzla is a solver for the theories of bit-vectors, floating-points, arrays and uninterpreted functions and their combinations (with and without quantifiers).

I. INTRODUCTION

Bitwuzla is a Satisfiability Modulo Theories (SMT) solver for the (quantifier-free and quantified) theories of fixed-size bit-vectors, floating-point arithmetic, arrays and uninterpreted functions and their combinations. A comprehensive system description of Bitwuzla can be found in [10].

Bitwuzla implements the lazy, abstraction/refinement-based SMT paradigm *lemmas on demand*, using a bit-vector abstraction similar to [6, 12].

Bit-Vectors. The bit-vector theory solver implements two orthogonal approaches, classic *bit-blasting* and *ternary propagation-based local search* [9], and a sequential combination of both. Additionally, Bitwuzla’s bit-blasting solver integrates an abstraction-refinement approach for bit-vector arithmetic as described in [11].

Arrays. The array theory solver implements and extends the array procedure from [6] with support for reasoning over (equalities of) nested arrays.

Floating-Point Arithmetic. For the theory of floating-point arithmetic, Bitwuzla implements a technique called *word-blasting* and integrates SymFPU [5], a C++ library of bit-vector encodings of floating-point operations.

Uninterpreted Functions. For uninterpreted functions, Bitwuzla implements a lazy form of Ackermann’s reduction called *dynamic Ackermannization* [7].

Quantifiers. Quantifier reasoning is handled by a dedicated quantifiers module, which implements model-based quantifier instantiation (MBQI) [8].

Bitwuzla further supports incremental solving (including incremental preprocessing), model generation, unsat core and unsat assumptions extraction.

This paper serves as system description for Bitwuzla as entered in the SMT competition 2026 [2]. Bitwuzla is licensed under the MIT license. Source code, releases and more information is available on the Bitwuzla website [1].

II. CONFIGURATIONS

Bitwuzla participates in the single query, incremental, unsat core, model validation, and parallel tracks in the logics matching the following regular expression:

$\wedge (\text{QF_}) ? (A) ? (UF) ? (BV | FP | FPLRA) + \$$

Bitwuzla uses CaDiCaL version 2.1.2 [4] as default SAT back end. Our abstraction-refinement approach [11] is enabled by default, with the minimum bit-vector size of arithmetic terms to abstract set to 33.

A. Improvements

- Full support for extensional constant arrays.
- Improved push/pop on the SAT level, now uses activation variables.
- Floating-point constants and constant folding are now based on the GNU MPFR library [3].

III. LICENSE

Bitwuzla is licensed under the MIT license. For more details, refer to the actual license text, which is distributed with the source code.

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